Assignment - 7

By NETHRA G

Register no: 240701358

Five topics, each with a short overview, 20 multiple-choice questions.

1. The Limits of Low-Code Platforms

Overview. Low-code platforms speed up delivery by replacing hand-written code with visual builders, prebuilt components and managed infrastructure. That abstraction has a cost. Typical limits are vendor lock-in (apps depend on a proprietary runtime), platform quotas on data, API calls and execution time, restricted UI customization, weaker version control and testing, and awkward expression of complex algorithms. Organizations also face governance risks such as shadow IT, security misconfiguration by non-technical builders, technical debt from undocumented workflows, and licensing costs that grow with users or apps. Most platforms offer escape hatches (custom code, APIs, widgets) and the usual mitigation is a hybrid design: low-code for workflows and forms, pro-code services for heavy logic.

Questions

1. What does “vendor lock-in” mean for a low-code application?

A) The vendor limits how many users can log in
B) The app is tightly tied to the platform’s proprietary runtime and is hard to move elsewhere
C) The vendor takes ownership of the customer’s data
D) The app works on only one device type

2. Which factor most commonly limits scalability on a low-code platform?

A) The number of colour themes available
B) Lack of drag-and-drop tools
C) Platform quotas on records, API calls and execution time
D) Browser cache size

3. “Shadow IT” in a low-code context refers to:

A) Apps built by business users outside IT oversight
B) Apps that run only at night
C) Backup servers hidden from users
D) Encrypted databases

4. Which is a typical limit when customizing a low-code user interface?

A) Unlimited pixel-level control
B) Direct access to the server operating system
C) Ability to compile custom binaries
D) Being restricted to the platform's components, themes and layout options

5. How does technical debt commonly build up in low-code apps?

A) Through excessive unit test coverage
B) Through poorly structured, undocumented workflows and quick fixes
C) Through using too many open standards
D) Through the platform's automatic updates only

6. Which of these is commonly harder in low-code than in traditional development?

A) Building a data entry form
B) Creating an approval workflow
C) Fine-grained version control, diffing and automated testing
D) Sending an email notification

7. Low-code is generally the best fit for:

A) Operating system kernels
B) Internal workflow and data-driven business apps
C) Game engines
D) Firmware for embedded devices

8. Which workload is a poor fit for a low-code platform?

A) Leave-request app
B) Inventory tracker
C) Customer onboarding portal
D) Ultra-low-latency, high-frequency trading engine

9. Which practice best addresses governance risk?

A) Banning all citizen developers
B) Letting every user publish apps freely
C) Defining roles, review processes and standards, often via a Center of Excellence
D) Disabling all integrations

10. Which pricing risk is common with low-code platforms?

A) Per-user or per-app licensing that grows costly at scale
B) Free unlimited usage forever
C) One-time perpetual licence with no renewal
D) Charges only for failed requests

11. A key security concern with citizen-built apps is:

A) Excessive encryption

B) Misconfigured permissions that expose sensitive data
C) Too many code reviews
D) Slow login pages

12. Why can low-code apps sometimes perform worse than hand-optimized code?

A) They never use databases
B) They always run offline
C) Generic abstraction layers add overhead and limit low-level tuning
D) They use too few workflows

13. What is an “escape hatch” in a low-code platform?

A) A refund policy
B) A backup restore feature
C) A user logout option
D) An extension point such as custom code, APIs or widgets that goes beyond the visual builder

14. Which integration limitation is typical?

A) Prebuilt connectors may not exist for legacy or niche systems, requiring custom API work
B) Platforms cannot connect to any external system
C) All integrations are free and unlimited
D) Integrations require physical cables

15. Which is a risk of depending heavily on a single platform?

A) Reduced need for documentation
B) Vendor changes to pricing, features or product direction can affect your apps
C) Automatic portability to all clouds
D) Guaranteed lifetime support

16. Debugging in low-code can be challenging mainly because:

A) Errors never occur
B) Code is always obfuscated by law
C) You have limited visibility into the underlying runtime and logs
D) Debuggers are banned

17. Which kind of logic is awkward to express in visual builders?

A) Simple field validation
B) Basic email triggers
C) Record creation
D) Complex algorithms with heavy computation or intricate data structures

18. Which of the following is NOT a limitation of low-code?

A) Faster delivery than hand-coding
B) Vendor dependency
C) Quotas and platform limits
D) Limited deep customization

19. Which action best reduces lock-in risk?

A) Avoiding documentation
B) Keeping business logic documented, using open APIs and planning data export
C) Embedding all logic in one hidden script
D) Never integrating with other tools

20. A common hybrid approach to overcome low-code limits is:

A) Rewrite everything in assembly
B) Avoid backend services completely
C) Use low-code for forms and workflows and pro-code services for heavy logic
D) Replace all users with bots

Answer key: 1-B, 2-C, 3-A, 4-D, 5-B, 6-C, 7-B, 8-D, 9-C, 10-A, 11-B, 12-C, 13-D, 14-A, 15-B, 16-C, 17-D, 18-A, 19-B, 20-C

2. Glimpses of Pro-Code in Zoho Creator and Catalyst

Overview. Zoho Creator is a low-code application platform that lets you go beyond drag-and-drop through **Deluge**, Zoho's scripting language for workflows, validations and integrations; **Widgets** built with HTML, CSS and JavaScript; **client functions**; **custom APIs** and REST APIs; and **Connections** for authenticated access to third-party services via invokeUrl. **Zoho Catalyst** is Zoho's serverless, full-stack pro-code platform. It offers **Functions** (Basic I/O, Advanced I/O, Event, Cron and others) in runtimes such as Java, Node.js and Python, a **Data Store** queried with **ZCQL**, **File Store**, **Cache**, **Authentication**, **AppSail** for managed app hosting, and the **Catalyst CLI** for local development and deployment. A common pattern is a Creator front end for forms and workflows calling Catalyst functions for heavy processing. Product features evolve, so check Zoho's current documentation for details.

Questions

1. Which scripting language is used in Zoho Creator?

A) Apex
B) Liquid
C) Deluge
D) Velocity

2. Zoho Catalyst is best described as:

A) A spreadsheet tool
B) A serverless full-stack platform for pro-code development
C) A no-code form builder only
D) An email client

3. Which Creator feature lets you embed custom HTML/CSS/JavaScript components?

A) Reports
B) Blueprints
C) Subforms
D) Widgets

4. Which Catalyst function type is meant for HTTP-style APIs with full control over request and response?

A) Advanced I/O functions
B) Cron functions
C) Event functions
D) Static functions

5. Which Catalyst function type runs on a schedule?

A) Browser Logic
B) Basic I/O
C) Cron functions
D) Advanced I/O

6. Which query language is used with the Catalyst Data Store?

A) GraphQL
B) ZCQL
C) SPARQL
D) XPath

7. Which Catalyst component stores files such as documents and images?

A) Cache
B) Cron
C) Circuits
D) File Store

8. Which Deluge task is used to call an external REST API from Creator?

A) invokeUrl
B) sendmail only
C) openUrl only
D) alert

9. In Zoho Creator, “Connections” are used to:

A) Link two forms visually
B) Print reports
C) Manage authenticated access to third-party services such as via OAuth
D) Connect to Wi-Fi

10. Catalyst AppSail is:

A) A drawing tool
B) A managed runtime for hosting applications built with stacks such as Node.js, Java or Python
C) A mobile keyboard
D) A billing module

11. Catalyst Event functions are generally:

A) Called only manually
B) Used only for UI styling
C) Used only for printing
D) Triggered by events from Catalyst components

12. How can external applications read and write Creator data programmatically?

A) Through Creator's REST APIs and custom APIs
B) Only through screenshots
C) Only by email
D) Not possible

13. Which runtimes can be used for Catalyst functions?

A) COBOL only
B) Swift only
C) Java, Node.js and Python
D) Fortran only

14. Catalyst Cache is:

A) A long-term archive
B) An in-memory key-value store for fast access
C) A video player
D) An email service

15. A Creator Custom API lets you:

A) Change the Creator logo
B) Delete the platform
C) Define your own endpoints backed by Deluge logic
D) Rename fields only

16. Which is a typical Creator and Catalyst integration pattern?

A) Catalyst renders Creator's forms
B) Creator replaces Catalyst's runtime
C) Both platforms must share one database engine
D) A Creator app calls a Catalyst function for heavy or custom processing

17. Catalyst Authentication provides:

A) Managed user sign-up and login for your apps
B) Image editing
C) Spreadsheet formulas
D) Video hosting

18. Catalyst Basic I/O functions are best suited to:

A) Long-running video rendering
B) Simple, lightweight tasks that take basic input and return output
C) Operating system drivers
D) Hosting databases

19. Creator Widgets are typically built with:

A) Only Deluge
B) Only SQL
C) HTML, CSS, JavaScript and the Creator JS SDK
D) Only Excel macros

20. The Catalyst CLI is used to:

A) Edit photos
B) Send newsletters
C) Manage payroll
D) Develop, test and deploy Catalyst projects from the command line

Answer key: 1-C, 2-B, 3-D, 4-A, 5-C, 6-B, 7-D, 8-A, 9-C, 10-B, 11-D, 12-A, 13-C, 14-B, 15-C, 16-D, 17-A, 18-B, 19-C, 20-D

3. The No-Code to Low-Code to Pro-Code Spectrum Revisited

Overview. Application development is best seen as a spectrum rather than three separate camps. **No-code** tools let business users assemble apps visually with no programming. **Low-code** tools add scripting and extensibility (for example Deluge in Zoho Creator) so builders can handle more complex needs. **Pro-code** means full traditional development with complete control (for example Zoho Catalyst functions written in Java, Node.js or Python). Moving right increases flexibility, control and required skill; moving left increases speed and accessibility. The boundaries are blurring: no-code tools gain scripting, pro-code tools gain visual and AI assistance, and **fusion teams** of business and IT staff mix approaches within one solution. Good

practice is to pick the right point on the spectrum for each component, weighing complexity, scalability, team skills, governance and total cost of ownership.

Questions

1. No-code platforms primarily target:

A) Business users who build apps visually without programming
B) Kernel developers
C) Database engine designers
D) Network engineers only

2. Low-code differs from no-code mainly because it:

A) Forbids visual tools
B) Adds scripting and extensibility on top of visual building
C) Requires assembly language
D) Cannot integrate with APIs

3. Pro-code refers to:

A) Apps built only by professionals' spouses
B) Apps with no logic
C) Traditional development with full control over code and infrastructure
D) Apps that cannot be deployed

4. A "citizen developer" is:

A) A government-employed programmer
B) A developer who only works remotely
C) A person who tests apps only
D) A business user who builds applications without formal programming training

5. Moving toward the pro-code end of the spectrum generally increases:

A) Control and flexibility, along with the skill required
B) Speed of first prototype
C) Accessibility to non-technical users
D) Platform handholding

6. Moving toward the no-code end generally increases:

A) Custom algorithm freedom
B) Infrastructure control
C) Speed of delivery and accessibility
D) Low-level performance tuning

7. A "fusion team" is:

A) A team that merges two companies

B) A mixed team of business and IT people building solutions together
C) A team of only database administrators
D) A hardware engineering group

8. Where does Deluge scripting in Zoho Creator typically sit on the spectrum?

A) Pure no-code
B) Machine code
C) Low-code scripting
D) Hardware design

9. Where do Zoho Catalyst serverless functions written in Java, Node.js or Python sit?

A) No-code
B) Spreadsheet formulas
C) Drag-and-drop only
D) Pro-code

10. Which factor is LEAST relevant when choosing a point on the spectrum?

A) Complexity of requirements
B) Team skills
C) Expected scalability
D) The colour of the company logo

11. A simple approval form for internal use is best suited to:

A) No-code or low-code
B) A custom operating system
C) A hand-built compiler
D) Embedded firmware

12. A custom machine-learning pipeline with bespoke infrastructure is best suited to:

A) A no-code form builder
B) A spreadsheet
C) Pro-code
D) A survey tool

13. Hybrid architectures:

A) Forbid mixing tools
B) Combine tools from different points on the spectrum in one solution
C) Use only no-code
D) Use only pro-code

14. Zoho Flow is best described as:

A) A relational database
B) A video editor

C) A serverless runtime
D) A no-code/low-code integration and automation tool

15. Why is the spectrum being “revisited”?

A) Because the terms were abolished
B) Because pro-code disappeared
C) Because boundaries are blurring as no-code gains scripting and pro-code gains visual and AI tooling
D) Because no-code failed completely

16. What can you do when an app outgrows its no-code tool?

A) Extend it with code or migrate parts of it
B) Delete the data
C) Stop using computers
D) Ignore the limits

17. A risk of choosing a point too far toward no-code is:

A) Excessive flexibility
B) Hitting a capability ceiling later
C) Needing too many developers
D) Slow login only

18. A risk of choosing a point too far toward pro-code is:

A) Too little control
B) Instant delivery
C) Slower delivery and higher build and maintenance cost
D) No need for testing

19. Total cost of ownership across the spectrum should consider:

A) Licence price only
B) Developer salary only
C) Hosting only
D) Build, maintenance, skills, licensing and infrastructure

20. Which is a best practice?

A) Match the tool to each component’s requirements rather than using one approach everywhere
B) Use one tool for everything regardless of fit
C) Avoid all documentation
D) Never reassess choices

Answer key: 1-A, 2-B, 3-C, 4-D, 5-A, 6-C, 7-B, 8-C, 9-D, 10-D, 11-A, 12-C, 13-B, 14-D, 15-C, 16-A, 17-B, 18-C, 19-D, 20-A

4. Career Pathways

Overview. Low-code and pro-code skills open several career paths. Entry points include **citizen developer**, **business analyst** and **low-code developer** (for example a Zoho Creator or Deluge developer). From there people move toward **solution architect**, **platform administrator**, **Center of Excellence lead** or into **pro-code roles** such as full-stack or serverless cloud developer (for example on Zoho Catalyst). Valuable complementary skills include data modeling, REST APIs, OAuth and webhooks, JavaScript, security and data privacy, and strong communication with stakeholders. Platform certifications and a portfolio of real, documented apps help employers judge ability. A "T-shaped" profile (broad knowledge, one deep specialty) and continuous learning are especially useful because platforms and AI tools change quickly.

Questions

1. Which role builds business applications professionally on a low-code platform?

A) Network cable installer
B) Low-code developer
C) Hardware assembler
D) Data centre guard

2. Deluge skills are most valuable for roles in:

A) The Zoho ecosystem, such as Creator developer roles
B) Game graphics only
C) Operating system design
D) Chip manufacturing

3. Which skill best complements low-code development?

A) Calligraphy
B) Typing speed only
C) Data modeling and query skills
D) Printer repair

4. A "T-shaped" professional has:

A) Skills in exactly two areas
B) Only deep skills in one narrow area
C) No specialty
D) Broad knowledge across areas and deep expertise in one

5. A Low-Code Center of Excellence role typically focuses on:

A) Governance, standards and enablement
B) Office catering

C) Hardware procurement
D) Logo design

6. A platform administrator mainly:

A) Writes only marketing copy
B) Manages users, permissions and environments
C) Designs microprocessors
D) Runs payroll

7. What advantage does a business analyst bring to low-code work?

A) Avoiding stakeholders
B) Writing only assembly
C) Translating business processes into sound app designs
D) Replacing all testing

8. A practical path from low-code to pro-code is to learn:

A) Only more drag-and-drop
B) Nothing new
C) Only office tools
D) JavaScript, APIs and serverless cloud development such as Catalyst

9. Platform certifications are useful because they:

A) Help validate platform skills to employers
B) Replace all experience
C) Guarantee a job
D) Are required by law

10. A solution architect typically:

A) Only fixes printers
B) Designs the overall integration, data and security approach across platforms
C) Only enters data
D) Only tests UI colours

11. Which soft skill is especially critical in low-code roles?

A) Avoiding feedback
B) Working in isolation
C) Clear communication with stakeholders
D) Ignoring requirements

12. Which best strengthens a portfolio?

A) Screenshots of other people's work
B) A blank resume
C) Random certificates only

D) Real, documented apps that solve real problems

13. How does low-code affect professional developers?

A) It shifts effort toward higher-value work rather than eliminating the need for them
B) It makes them illegal
C) It removes all demand
D) It bans coding

14. Which skill is increasingly important alongside coding?

A) Ignoring AI
B) Using AI assistants effectively and validating their output
C) Avoiding documentation
D) Manual memory management only

15. A citizen developer can grow into:

A) Only a retired role
B) Nothing else
C) A formal developer, product owner or analyst
D) A hardware vendor only

16. Freelancing with low-code is attractive because it allows:

A) Rapid delivery of business apps for small and medium businesses
B) No client contact
C) Zero learning
D) Guaranteed fixed income

17. Which knowledge is essential for every builder?

A) Only graphic design
B) Only hardware repair
C) Only spreadsheets
D) Security and data privacy

18. Integration skills commonly include:

A) Cable crimping
B) REST APIs, webhooks and OAuth
C) Typewriter repair
D) Photography

19. Why is continuous learning vital in this field?

A) Platforms never change
B) Learning is discouraged
C) Platforms and AI tooling evolve quickly
D) Skills are permanent once learned

20. Pro-code work on a platform like Catalyst can lead to a role as:

A) Full-stack or serverless cloud developer
B) Office receptionist
C) Warehouse picker
D) Graphic printer

Answer key: 1-B, 2-A, 3-C, 4-D, 5-A, 6-B, 7-C, 8-D, 9-A, 10-B, 11-C, 12-D, 13-A, 14-B, 15-C, 16-A, 17-D, 18-B, 19-C, 20-A

5. The Future of Low-Code

Overview. Analysts expect low-code and no-code to account for a growing share of new application development. The biggest shift is generative AI: builders can describe an app in natural language (sometimes called "vibe coding") and have the platform generate data models, workflows and code. Related trends include **hyperautomation** (RPA, AI and low-code combined), **agentic AI** (autonomous agents acting through platform APIs), **composable** architectures built from reusable components, **multi-experience** apps across web, mobile, chat and voice, and convergence between low-code and pro-code tooling. As more apps are created by more people, **governance, security, testing and compliance** become more important, not less, and AI-generated output still needs human review. Professional developers increasingly act as architects, reviewers and orchestrators, while open standards and APIs help limit lock-in.

Questions

1. What do analysts generally expect for low-code and no-code?

A) They will disappear soon
B) They will account for a growing share of new application development
C) They will be limited to games
D) They will replace all databases

2. Generative AI is changing low-code mainly by enabling:

A) App creation from natural-language descriptions
B) Removal of all data
C) Manual-only workflows
D) Paper-based forms

3. "Vibe coding" refers to:

A) Coding while listening to music only
B) Writing code in assembly
C) Describing intent in natural language and letting AI generate the code
D) Avoiding all testing

4. Hyperautomation combines:

A) Only spreadsheets
B) Only email
C) Only manual work
D) RPA, AI and low-code to automate processes end to end

5. A composable approach means:

A) Assembling solutions from modular, reusable components
B) Writing one giant monolith
C) Avoiding APIs
D) Using only one vendor's logo

6. As AI-generated apps multiply, governance will likely:

A) Become unnecessary
B) Become more important
C) Be handled by chance
D) Be banned

7. Agentic AI refers to:

A) Insurance agents
B) Travel agents
C) AI agents that autonomously carry out tasks and workflows
D) Agents in spreadsheets only

8. Convergence in this context means:

A) Pro-code developers refusing visual tools
B) No-code adding nothing new
C) Platforms becoming identical in price
D) Low-code and pro-code tooling moving closer, with developers also using low-code

9. Zoho's Zia is:

A) Zoho's AI assistant spanning its products
B) A database engine
C) A billing plan
D) A hardware device

10. A key risk of AI-generated apps is:

A) They are always perfect
B) Hidden bugs or security flaws if not reviewed
C) They never run
D) They are too well documented

11. The developer's role is likely to shift toward:

A) Typing boilerplate only
B) Avoiding platforms
C) Architecting, reviewing and orchestrating
D) Manual deployment only

12. Open standards and APIs help by:

A) Increasing lock-in
B) Blocking integration
C) Hiding data
D) Reducing lock-in and easing integration

13. Multi-experience development means building apps for:

A) Only desktop browsers
B) Web, mobile, chat and voice channels
C) Only printers
D) Only email

14. The citizen developer population is likely to:

A) Shrink to zero
B) Stay exactly the same
C) Grow, supported by AI and guardrails
D) Be replaced by compilers

15. Data privacy regulation will most likely:

A) Be ignored by platforms
B) Apply only to hardware
C) End all cloud use
D) Require compliance features built into platforms

16. Why does an API-first design matter for AI agents?

A) It lets agents and tools operate platform functions programmatically
B) It blocks automation
C) It removes the need for security
D) It makes apps offline-only

17. Which shift describes the direction of development tools?

A) From outcomes to manual screens
B) From building screens by hand toward specifying intent and outcomes
C) From the cloud to paper
D) From APIs to fax

18. Pro-code will likely remain vital for:

A) Nothing

B) Simple forms only
C) Scale, performance and novel capabilities
D) Spreadsheets only

19. How might AI affect cost?

A) Maintenance disappears completely
B) Build cost cannot change
C) Testing becomes obsolete
D) Build cost may fall, while maintenance, security and governance costs remain

20. The best way to prepare for this future is to:

A) Learn fundamentals, AI-assisted tools and governance practices
B) Avoid all new tools
C) Memorize one platform's menus only
D) Ignore security

Answer key: 1-B, 2-A, 3-C, 4-D, 5-A, 6-B, 7-C, 8-D, 9-A, 10-B, 11-C, 12-D, 13-B, 14-C, 15-D, 16-A, 17-B, 18-C, 19-D, 20-A